

Element K: Reflection on the Design Project

Project Overview

As a team, we feel that we have created and documented a viable, exciting, and advanced-level solution with the time and resources we were given. At the end of our time for this project, we have constructed a rudimentary prototype and created a more developed CAD in Fusion 360.

Major Steps

EDP Step (Iteration #)

Ask I - During this initial phase of the project, we asked our client and primary stakeholder, Mr. Kiser, in-depth questions. We were able to identify key constraints (timespan, resources, ergonomics, and possible load on the cart) and criteria (comfort, durability, compactness, removability). The success of this stage allowed us to succeed in the later stages of the EDP.

Research/Brainstorm I - Initially, we considered various currently existing solutions, ranging from Walmart products to a patented design for an electric office chair. By evaluating each of these designs, we were able to generate 8+ concepts per group member. Then, we chose the best/most viable concepts to move forward with. We successfully generated a wide range of solutions, with varying degrees of plausibility and creativity, but had a slight bias towards the electric pulling pod for its novelty and distinct upsides. This bias skewed the rest of our project, especially during the initial Select phase.

Select I - Next, we created decision matrices based on the criteria identified during the Ask stage, then weighted the criteria based on relative importance to our primary stakeholders. Each member of the group then went through and scored each of the solutions for each of the criteria categories 1-10. With all the weighted scores, we determined that the electric pulling pod was the strongest solution.

Prototype I - After finding a solution, we initially used Cole's VEX robot to get a general feel for how much power our robot would need to pull the cart. After finding success with the 6-motor, 66 total-watt drive, we determined that we could move forward with a similarly powered solution.

Prototype II - We then approached creating a physical prototype, using wood, PVC, and caster wheels. This prototype allowed us to test different hook designs, especially around hook geometry and a latching system. Due to the more complex mechanical and electrical nature of our solution, this solution couldn't fully represent all elements of our solution. However, this step was still successful, as it allowed us to consider the importance of the center of mass and our connection point to the cart.

Prototype III - After considering our initial prototypes, we created a CAD in Fusion 360 that was much more accurate to our solution, with a full chassis, microcontroller, motors, battery, and wheels. This was a super important prototype, allowing us to properly present our ideas to the experts.

Test III - We spent a day presenting our prototypes II and III to a wide range of engineering-related experts from across Nashville. We received a variety of feedback, ranging from the weight of our robot to the user experience of remote controlling the pod vs it autonomously following the user. This was a huge success for us, as it acted as validation for our current plans and a rich source of information for improvements.

Improve III - After taking into consideration the feedback we got from the industry experts, we were able to further improve our CAD from Prototype IV. Improvements included adding a passive rubber-band latch on the hooks, creating a shorter arm between the top of the chassis and the hooks (to reduce torque on the hook arms),

Lessons Learned

- 1) Make sure you have the budget - we severely underestimated the budget requirements of the project, especially when the motor performance starts exceeding the N10 gearmotor range. Even with a \$30 materials budget and a slew of personal inventory, we still had trouble sourcing and purchasing the parts necessary to properly prototype our solution.
- 2) Understand the physics and consult your teacher/mentor - We didn't fully grasp the role of device weight in being able to pull the cart until after the expert review day and the subsequent documentation. If I could do the project again, I would complete Element E soon after settling on a solution and get it checked by someone with a solid physics background. This way, the team can fully grasp the factors that play into this sort of project.